

Vitwerlijnen §3.8

$$1. a. s \cdot Y(s) + 4 Y(s) = 6X(s) \Leftrightarrow (s+4) \cdot Y(s) = 6 \cdot X(s)$$

$$\Leftrightarrow H(s) = \frac{Y(s)}{X(s)} = \frac{6}{s+4}$$

$$b. s^2 Y(s) + 6s Y(s) + 0 Y(s) = 3s X(s) + 4 X(s) \Leftrightarrow$$

$$(s^2 + 6s + 0) \cdot Y(s) = (3s + 4) \cdot X(s) \Leftrightarrow$$

$$H(s) = \frac{Y(s)}{X(s)} = \frac{3s+4}{s^2+6s+0}$$

$$c. 4s^2 \cdot Y(s) + 16s Y(s) + 32 Y(s) = 2X(s) \cdot s + X(s) \Leftrightarrow$$

$$(4s^2 + 16s + 32) \cdot Y(s) = (2s + 1) X(s) \Leftrightarrow$$

$$\frac{Y(s)}{X(s)} = \frac{2s+1}{4s^2+16s+32}$$

$$d. s^3 \cdot Y(s) + 3s^2 Y(s) + 3s Y(s) + Y(s) = X(s) \Leftrightarrow$$

$$(s^3 + 3s^2 + 3s + 1) \cdot Y(s) = X(s) \Leftrightarrow$$

$$H(s) = \frac{Y(s)}{X(s)} = \frac{1}{s^3+3s^2+3s+1}$$

$$e. s^2 Y(s) + 3s Y(s) + 2,25 Y(s) = s X(s) + 4 X(s) \Leftrightarrow$$

$$(s^2 + 3s + 2,25) Y(s) = (s + 4) \cdot X(s) \Leftrightarrow$$

$$\frac{Y(s)}{X(s)} = \frac{s+4}{s^2+3s+2,25}$$

$$f \quad s^3 Y(s) + 4s^2 Y(s) + 3s Y(s) + 12 Y(s) = X(s) \Leftrightarrow$$

$$(s^3 + 4s^2 + 3s + 12) Y(s) = X(s) \Leftrightarrow$$

$$\frac{Y(s)}{X(s)} = \frac{1}{s^3 + 4s^2 + 3s + 12}$$

2 impuls, dus $x(t) = \delta(t)$ en $X(s) = 1$. Omdat $Y(s) = X(s) \cdot H(s)$ volgt hier dus dat $Y(s) = 1 \cdot H(s) = H(s)$

$$a \quad y(t) = \mathcal{L}^{-1}\left(\frac{6}{s+4}\right) = ~~6e^{-4t}~~ 6 \cdot e^{-4t}$$

$$b \quad y(t) = \mathcal{L}^{-1}\left(\frac{3s+4}{s^2+6s+8}\right) = \mathcal{L}^{-1}\left(\frac{4}{s+4} - \frac{1}{s+2}\right) = 4e^{-4t} - e^{-2t}$$

$$c \quad y(t) = \mathcal{L}^{-1}\left(\frac{2s+1}{4s^2+6s+32}\right) = \frac{1}{2} \mathcal{L}^{-1}\left(\frac{s+1/2}{s^2+3/2s+8}\right)$$

$$= \frac{1}{2} \mathcal{L}^{-1}\left(\frac{s+3/4 - 1/4}{(s+3/4)^2 + \frac{119}{16}}\right) = \frac{1}{2} \mathcal{L}^{-1}\left(\frac{s+3/4}{(s+3/4)^2 + \frac{119}{16}} - \frac{1/4}{(s+3/4)^2 + \frac{119}{16}}\right)$$

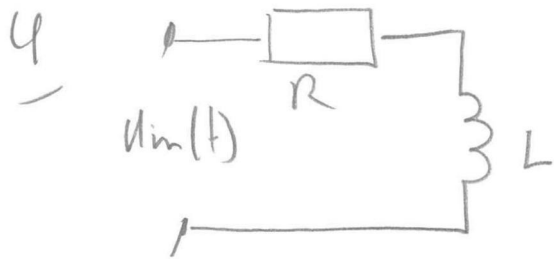
$$= \frac{1}{2} \left(e^{-3/4 t} \cdot \cos\left(\frac{1}{4} \sqrt{119} \cdot t\right) - \frac{1}{4} \cdot \frac{4}{\sqrt{119}} \cdot \sin\left(\frac{1}{4} \sqrt{119} \cdot t\right) \right)$$

2c: Zie aparte uitwerking

3. Einheitsimpuls: $x(t) = 1$, das $X(s) = \frac{1}{s}$. $Y(s) = X(s) \cdot H(s)$
 $= \frac{1}{s} \cdot H(s)$. Einheitsimpulsantwort wird das geben dann:
 $y(t) = \mathcal{L}^{-1}\left(\frac{1}{s} \cdot H(s)\right)$

a. $y(t) = \mathcal{L}^{-1}\left(\frac{1}{s} \cdot \frac{6}{s+4}\right) = \mathcal{L}^{-1}\left(\frac{3/2}{s} - \frac{3/2}{s+4}\right)$
 $= 3/2 - 3/2 e^{-4t}$

b. $y(t) = \mathcal{L}^{-1}\left(\frac{1}{s} \cdot \frac{3s+4}{s^2+6s+8}\right) = \mathcal{L}^{-1}\left(\frac{1/2}{s} + \frac{1/2}{s+2} - \frac{1}{s+4}\right)$
 $= 1/2 + 1/2 e^{-2t} - e^{-4t}$



$$u_{L(t)} = L \cdot i'_{L(t)}$$

$$u_{in}(t) = u_R(t) + u_L(t) = i(t)R + u_L(t) \Leftrightarrow$$

$$u'_{in}(t) = i'(t) \cdot R + u'_L(t) \Leftrightarrow u'_{in}(t) = \frac{R}{L} \cdot u_L(t) + u'_L(t)$$

$$\begin{aligned} \Leftrightarrow s \cdot u_{in}(s) &= \frac{R}{L} u_L(s) + s \cdot u_L(s) \\ &= \left(\frac{R}{L} + s \right) \cdot u_L(s) \end{aligned}$$

$$\Leftrightarrow \frac{u_L(s)}{u_{in}(s)} = \frac{s}{\frac{R}{L} + s} = \frac{L \cdot s}{R + L \cdot s}$$

$$\begin{aligned} u_{in}(t) &= i(t) \cdot R + u_L(t) \\ &= i(t) \cdot R + L \cdot i'_L(t) \quad \Leftrightarrow \end{aligned}$$

$$\begin{aligned} u_{in}(s) &= i(s) \cdot R + L \cdot s \cdot i(s) \\ &= (R + L \cdot s) i(s) \quad \Leftrightarrow \end{aligned}$$

$$\frac{i(s)}{u_{in}(s)} = \frac{1}{R + L \cdot s}$$